

# The Importance of Penetration Testing

Penetration testing is an established technology for discovering vulnerabilities, threats and risks, as well as offering solutions to malware attacks on Web applications by malicious players and hackers. Such testing is important because a successful attack can cause tremendous losses to enterprises and users of the application.



**S**ecurity testing has become a crucial aspect of the quality assurance process. Given the high number of malicious actors lurking in cyberspace with an eye on exploiting software vulnerabilities, QA teams must be driven about identifying and addressing such weak points. Penetration testing is perhaps the most helpful form of security testing when it comes to discovering and plugging holes in an app's structure and defence. Pen testing (penetration testing or security testing) is the method of testing your application for vulnerabilities.

## What is penetration testing?

The main objective of penetration testing is to identify security vulnerabilities, threats and risks in a

software application, network or Web application that an attacker could exploit. In simple words, the main idea of this test is to secure vital data from outsiders like hackers, who can gain illegal access to the application and exploit the app to access sensitive information if any kind of vulnerability is recognised within it. Penetration tests are also occasionally called white hat attacks.

## Types of penetration testing

The type of penetration test selected depends on the purpose and scope of the organisation's work, and whether it wants to simulate an attack by an employee, a network admin, or by external sources. Generally, there are three different

types of penetration testing:

- Black box testing
- White box testing
- Grey box testing

In the case of black box penetration testing, the testers are not provided with much knowledge about the application. They carry out tests and it is their responsibility to collect information about the target network, system, or application.

In white box penetration testing, the tester gets all the information about the network, system or application, along with the source code, OS details, etc. It can be considered a simulation of an attack by internal sources.

In grey box penetration testing, the tester has partial knowledge about the application or system. Thus, it can